

Yashveer Jain

[yashveerjain.github.io](https://github.com/yashveerjain) | [linkedin.com/in/yashveerjain/](https://www.linkedin.com/in/yashveerjain/) | yashveer@terpmail.umd.edu

SKILLS AND LANGUAGES

Machine Learning: TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV, Scikit-learn, Huggingface, TensorRT, Computer Vision, NLP, LLMs, VLMs, Langchain, Ollama.

Programming Languages: Python, C, C++, Cuda, Flutter, Dart, SQL

Development Tools: Git, Docker, Agile Development.

Cloud Platforms: AWS (Certified Cloud Practitioner), Azure, GCP.

Others: Rest API, Data structures, Algorithms, CUDA, Motion Planning, Graph Theory (DFS/BFS/A*),

WORK EXPERIENCE

Embedded AI Intern - Renesas Electronics, USA

May 2023–Aug 2023

- *Trained and deployed SVM and logistic regression models on Renesas RA6 Series and RL78 microcontroller, achieving 95% accuracy for an Asset Tracking Project, enabling efficient asset management.*
- *Ensured success of edge computing projects by developing and implementing machine learning models optimized for resource-constrained embedded systems.*

Machine learning Engineer - AIMonk Labs, India

Aug 2020–Aug 2022

- *Developed and deployed a robust testing infrastructure for AI models, integrating FastAPI backend on AWS EC2 instance, streamlining the model development lifecycle.*
- *Optimized OCR algorithms using Transformers implemented on PyTorch, achieving a 60% accuracy boost, significantly improving text recognition capabilities.*
- *Enhanced car detection AI by 20% using DeepLabv3 and YOLOv7 integration, leading to more accurate object detection in autonomous systems.*
- *Collaborated with cross-functional teams on Azure DevOps for model deployment on AWS SageMaker, ensuring seamless integration and deployment of machine learning models.*

Research Intern - Swaayatt Robotics, India

Jan 2020–May 2020

- *Implemented Kalman and particle filters with LK optical flow using OpenCV for autonomous vehicle and pedestrian tracking in autonomous driving systems increasing the detection accuracy by 20%.*

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Maryland (Fall 2022-Spring 2023)

Aug 2022-May 2023

TinyDepth: Generalized Neural Metric Depth for Palm-sized Robots. Under Review Nature npj Robotics.

Designed deep learning architecture to run on sub-6cm microdrone with embedded Raspberry Pi for autonomous navigation (Perception & Robotics Group, Prof. Yiannis Aloimonos).

PROJECTS

Automatic Labeling Mail - [GitHub](#)

May 2024-June 2024

This project uses the Gmail API and the llama3 based chatbot running locally to automatically label emails based on status (e.g., rejected, need action, applied) and generates a CSV file for better organization.

SVM-based Image Classification with Multithreading (C++)- [GitHub](#)

Mar 2024-May 2024

Implemented a Support Vector Machine (SVM) for image classification in C++. Utilized multithreading for improved performance on the CIFAR-10 dataset.

Home Automation - [GitHub](#)

Jun 2021-Aug 2021

Established a home server for device connectivity using FastAPI, MQTT, Arduino, Raspberry Pi. Demonstrating proficiency in backend development, hardware integration and IoT systems.

Home Automation App- [GitHub](#)

April 2021-Aug 2021

Developed a home automation app using Flutter, Dart, Firebase, and SQLite with real-time database, dynamic UI, and device management for controlling electrical appliances. Enhanced with room/device management, status updates, and time-based greetings.

UnDeepVO (Unsupervised Deep Learning Visual Odometry) - [GitHub](#)

Jan 2020 - May 2020

Implemented unsupervised deep learning visual odometry paper using python and tensorflow, to handle depth estimation and transformation (translation and rotation of camera) using monocular camera images.

EDUCATION

Master of Engineering (Robotics) (GPA:3.87/4)

Aug 2022–May 2024

University of Maryland – College Park (UMD), USA

Robot Perception, Becoming a Design Thinker, NLP, Visual Learning & Recognition, Deep Learning.

Bachelor of Engineering (Mechatronics) (GPA:9.44/10)

July 2016- July 2020

Manipal University Jaipur, Rajasthan, India

CERTIFICATE

CUDA programming Masterclass with C++ (Udemy)

Mar 2024

Gained knowledge in implementing CUDA algorithms and managing memory, including shared memory and global memory, as well as handling synchronization using streams.